

AI Practitioner Skills Framework

Print edition - v5.2 - August 27, 2026 - 7 core domains - License CC-BY-SA 4.0

01 - Technical Prompt Engineering

Constructing prompts as structured blueprints, not keyword lists. The input layer for all AI generation.

SUB-SKILL	EXAMPLE
Scaffold Method	A woman walking in rain + cinematic lighting + 85mm f/1.8 + photorealistic + 4K
Information Priority (Front-Loading)	Oil painting of a castle... (style first) vs. A castle, oil painting...
Photographic Vocabulary	85mm lens, f/2.8, shallow depth of field instead of hyperrealistic
Active Voice Editing	remove the background, add a red hat, turn the car into a truck
Agent Prompting Patterns	You are a cinematographer. First analyze the scene mood, then specify lighting.
Long-Horizon Video Scripts	Scene 1: Close-up on protagonist, golden hour. Transition: Cross-dissolve to wide establishing
Camera Movement Language	pull from close-up to wide shot, circular tracking shot, dolly zoom
Advanced Prompt Syntax	(masterpiece:1.5), (best quality:1.3), [cinematic lighting] :: 0.8
Semantic Layering	A [serious executive] [in shadows] [dramatic side lighting] creates different emphasis than [dr
Style Injection Techniques	::0.3 after style reference to preserve subject while adding aesthetic influence

02 - Photographic Literacy

Reconstructing real-world physics to achieve believable, professional-grade results.

SUB-SKILL	EXAMPLE
Lighting Pattern Mastery	Rembrandt lighting: key light 45deg high, fill at 1/4 power, subtle rim for separation
Lens Selection	85mm portrait lens for flattering facial compression and background bokeh
Focal Length Mapping	24mm at 2m = dramatic wide-angle distortion; 135mm at 5m = flattering compression with natural
Optical Physics for AI	Prompt: shot on Leica 50mm Summilux, visible anamorphic horizontal flares, organic bokeh circle
Aperture Control	f/2.8 for subject isolation; f/11 for product shots requiring full sharpness
Native Resolution Management	Generate at 4K native to capture skin pores and fabric weave; avoid upscaling artifacts
Vertical Format Mastery	Center subject with negative space above for TikTok text overlays; use leading lines vertically
Advanced Rendering Terms	subsurface scattering on skin/ears; ambient occlusion in fabric folds and crevices
Anamorphic Mastery	Prompt: shot on Panavision Primo anamorphic, horizontal blue lens flares, 2.39:1 aspect ratio,
Cinematic Grammar	Open on wide establishing shot, push in to medium, cut to close-up reaction; parallel editing a
Executive Portraiture	Prompt: CEO portrait, corner lighting establishing authority, dark navy suit against library ba
Quiet Luxury Aesthetic	Prompt: quiet luxury editorial, cream cashmere sweater, matte gold jewelry, no visible logos, s

03 - Strategic Negation & Skin Mastery

Telling the AI what not to include to overcome unnatural perfection and maintain visual integrity.

SUB-SKILL	EXAMPLE
Negative Prompting with Weights	(plastic skin:1.4), (cartoon:1.3), (extra fingers:1.4)
Skin Realism Management	Prompt: visible pores, fine vellus hair, subtle skin variation Negate: airbrushed, doll-like, w
Anatomical Correction	(fused fingers, extra limbs, double iris, asymmetrical eyes:1.4)
Temporal Consistency for Video	Use --cref with character sheet + retraining by error recycling for stable sequences
Drift Correction	Generate 10-frame clip, identify drift points, retrain prompt weights, regenerate with correcti

04 - Identity Preservation & Consistency

Maintaining specific characters, styles, and narratives across multiple generations and scenes.

SUB-SKILL	EXAMPLE
Seed Locking	--seed 12345 to reproduce base composition while varying style modifiers or lighting
Reference Tools	--cref character_sheet.jpg --sref art_style.jpg --cw 80
Character Weight (--cw)	--cw 80 to preserve face and clothing but allow pose variation and expression changes
Multi-Reference Consistency	char1.jpg + char2.jpg + background.jpg + prop.jpg for ensemble scene with consistent elements
Agent-Based Consistency	DirectorAgent coordinates Veo 3.1 + Midjourney for cross-scene visual continuity and script fid
Visual-Script Alignment (VSA)	Score generated frames against storyboard; flag deviations >15% for review and regeneration

05 - Post-Processing & Hybrid Workflows

Treating AI generation as the start of an iterative, multi-tool creative process -- not the final step.

SUB-SKILL	EXAMPLE
Iterative Refinement (Branching)	Keep --seed 12345; change weather: rainy to snowy; compare outputs side-by-side
Inpainting & Outpainting	Fix malformed hands via targeted inpainting; extend background via intelligent outpainting
External AI Enhancement	Upscale 4K native gen to 8K with Topaz; color grade in Lightroom for cinematic LUT application
Unified Generation-Editing	Iterate on image + text overlay + composition adjustments in single interface without context s
Cross-Domain Compositing	Composite illustrated character into photoreal background via precise masked inpainting and lig
Text Overlay Integration	Prompt: text: "LAUNCH 2026" in bold serif, bottom-center, gold foil texture with subtle drop sh
Post-Generation Mastery Checklist	Check hands/feet anatomy, verify eye direction consistency, confirm lighting coherence, test at

SUB-SKILL	EXAMPLE
Model Serving	REST/gRPC endpoints, Kubernetes orchestration, load balancers, auto-scaling groups
Quantization & Optimization	ONNX export, TensorRT, GGUF quantization, pruning, distillation techniques
Monitoring & Logging	Prometheus/Grafana dashboards, custom telemetry, LangSmith, structured logging
CI/CD for AI Pipelines	GitHub Actions, MLflow, model versioning, automated regression testing, canary deployments

06 - AI Agent Orchestration

Designing and managing multi-step, autonomous workflows where AI agents collaborate to achieve complex creative and operational goals.

SUB-SKILL	EXAMPLE
Agent Architecture Design	Agent workflow: search web, summarize findings, generate image, draft social post, schedule pub
Router Pattern Implementation	Query "create video ad for product launch" routed to VideoAgent; "write blog post" routed to Te
SLM Optimization	Use 7B SLM for email drafting and summarization; reserve 70B model for complex strategy and cre
Role-Based Agent Teams	PlannerAgent breaks brief into tasks, ResearcherAgent gathers references, WriterAgent drafts, R
Cross-Agent Communication	Agents publish/subscribe to task_context channel; use consensus voting for creative decisions
Autonomous Research Workflows	ResearchAgent tasked: "Gather Q1 2026 AI video generation trends from 5+ sources, output markdo
Private Skill Deployment	compliance-check.skill validates all outputs against company brand guidelines, legal requiremen
Auditable Agent Workflows	Log: Agent X called tool Y at Z time with input A, output B, human approval C, final deployment
Sovereign AI Infrastructure	Run agent cluster in private cloud; all data stays within organizational boundaries; zero exter
Skills Packaging (SKILL.md)	This document is a SKILL.md example; agents load only relevant sections per task via progressiv

07 - LoRA Ecosystem & Parameter-Efficient Fine-Tuning

Bridging prompt engineering and full model ownership through efficient adaptation techniques. LoRA enables targeted customization of foundation models for specific styles, subjects, and workflows without retraining the entire model.

SUB-SKILL	EXAMPLE
LoRA Fundamentals	<code>r=16, lora_alpha=32, target_modules=["to_q","to_k","to_v"]</code>
QLoRA Optimization	Fine-tune 7B model on single 12GB GPU with 4-bit quantization + LoRA rank 16
DoRA (Weight-Decomposed)	<code>decompose=True</code> in LoRA config for detail-sensitive portrait style adaptation
AdaLoRA (Adaptive Rank)	<code>rank_budget=0.5</code> - auto-allocates higher rank to critical attention layers
Rank-Stabilized LoRA (rsLoRA)	<code>rscale=True</code> - enables rank 64+ without training instability
LoRA+ (Differential LR)	<code>lr_A=1e-4, lr_B=1e-3</code> - B matrix learns 10x faster for better optimization
VeRA (Vector-based)	Train only 0.01% of parameters using shared frozen random bases
GLoRA (Global Prompt)	<code>task_descriptor</code> -> <code>global_adapter</code> -> <code>per_layer_lora</code> for style switching
LoRA Style Libraries	<code>style_noir@0.6 + lighting_rembbrandt@0.4 + subject_ceo@0.8</code>
LoRA Training Pipeline	20-50 curated images -> QLoRA rank 16 -> 10-20 epochs -> safetensors export
S-LoRA Serving	Load 50+ LoRA adapters dynamically per request with KV cache sharing
LoRA x Framework Integration	Character LoRA x style LoRA x lighting LoRA combined for brand campaign consistency

Platform & Tool Reference

Pro Image

Tools: Flux, Ideogram, Recraft, Qwen Image 2.0

Video Generation

Tools: Veo 3.1, Wan 2.6, Sora, Runway Gen-3

Hybrid Editing

Tools: Photoshop (Gen Fill), ComfyUI, Topaz Photo AI, ACE++

Agent Frameworks

Tools: LangChain, AutoGen, CrewAI, custom builders, SKILL.md parsers

Deployment

Tools: vLLM, TensorRT, ONNX Runtime, GGUF, Kubernetes

Monitoring

Tools: Prometheus/Grafana, LangSmith, Arize, custom telemetry

Resources

Interactive site

<https://marktantongco.github.io/aiskills-photog/>

Print wiki (HTML)

<https://marktantongco.github.io/aiskills-photog/wiki.html>

Source document

<https://marktantongco.github.io/aiskills-photog/SKILL.md>